

MCMC3001 ENVIRONMENTAL ENGINEERING (3-0-0)

Module-III: (08 Hrs)

Air - Composition and properties of air, Quantification of air pollutants, monitoring of air pollutants, Air pollution- Occupational hazards, Urban air pollution automobile pollution, Chemistry of combustion, Automobile engines, quality of fuel, operating conditions and interrelationship. Air quality standards, Control measures for Air pollution, construction and limitations.

Module -3

Air Pollution

Definition of Air pollution: It is the presence of substances in air in sufficient concentration and for sufficient time, so as to be, or threaten to be injurious to human, plant or animal life, or to property, or which reasonably interferes with the comfortable enjoyment of life and property.

The air Act of Govt. of India (amendment 1987) defines air pollution as “air pollution means any solid, liquid or gaseous substances present in the atmosphere in such concentrations that may tend to be injurious to human beings or other living creatures or plants or property or enjoyment”.

<i>Component</i>	<i>By volume</i>	<i>By weight</i>
Nitrogen	78.084%	75.51%
Oxygen	20.946%	23.15%
Argon	0.934%	1.28%
Carbon dioxide	0.033%	0.046%
Neon	18.180 ppm	12.50 ppm
Helium	5.240 ppm	0.72 ppm
Krypton	1.190 ppm	2.90 ppm
Xenon	0.087 ppm	0.36 ppm
Nitrous oxide	0.500 ppm	1.50 ppm
Methane	2.0 ppm	1.2 ppm
Hydrogen	0.5 ppm	0.03 ppm
Ozone	0.01 ppm	

Definition of Air pollutants:

Substances introduced into the air, natural or manmade, in concentrations detrimental to human, plant or animal life, or to property.

Major Classification of Air Pollutants:

- 1] Primary – Secondary
- 2] Natural – Manmade
- 3] Criteria Air Pollutants
- 4] Physical - chemical - biological

Primary pollutants and secondary pollutants: Primary pollutants are substances that are directly emitted into the atmosphere from sources. Primary pollutants are those that are emitted directly from identifiable sources. Secondary air pollutants are those that are produced in the air by the interaction of two or more primary air pollutants.

Primary Air pollutants:-

- (i) Fine (less than 100μ) and coarse (more than 100μ) suspended particulate matter
- (ii) Oxides of sulfur
- (iii) Oxides of nitrogen
- (iv) Carbon monoxide

- (v) Halogens
- (vi) Organic compounds
- (vii) Radioactive compounds

Secondary Air pollutants:-

- (i) Ozone
- (ii) PAN (peroxy acetyl nitrate)
- (iii) Photo chemical smog
- (iv) Acid mists

Air pollutants arise from both manmade and natural processes. The ambient air quality may be defined by the concentration of a set of pollutants which may be present in the ambient air we breathe in. These pollutants may be called **criteria pollutants**.

Natural Contaminants: Pollen is important natural contaminant because of its peculiar properties of irritation and allergy sometimes leading to bronchitis, asthma and dermatitis. Pollen grains are the male gametophytes of gymnosperms and angiosperms and they are discharged into the atmosphere from plants etc. The air transported pollen grains range mainly between 10 and 50 microns. Manmade refers to any pollutant produced to influence or action of humans.

Aerosols: Aerosols refer to the dispersion of solid or liquid particles of microscopic size in the air. It can also be defined as a colloidal system in which the dispersion medium is gas and the dispersed phase is solid or liquid. The term aerosol is applicable until it is in suspension and after settlement due to its own weight or by addition with other particles (agglomeration) it is no longer an air pollutant. The diameter of the aerosol may range from 0.01 (or less) micron to 100 micron.

The various aerosols are as follows:-

- (i) **Dust:** Dust is produced by the crushing, grinding and natural sources like windstorms. Generally the dust particles are over 20 micron in diameter. They do not flocculate but settle under gravity, but smaller particles like 5 micron form stable suspensions.
- (ii) **Smoke:** Smoke is made up of finely divided particles produced by incomplete combustion. Generally it consists of carbon particles of size less than 1.0 micron.
- (iii) **Mists:** Mist is a light dispersion of minute water droplets suspended in the atmosphere ranging from 40 to 400 micron in size.

(iv) **Fog:** Fog is made up of dispersion of water or ice near the earth's surface reducing visibility to less than 500 m. In natural fog the size of particles range from 1.0 to 40 micron.

(v) **Fumes:** Fumes are solid particles generated by condensation from the gaseous state after volatilization from melted substances. Fumes flocculate and sometimes coalesce.

Gases: Following are the **main air pollutant gases**

(i) **Sulphur dioxide:** It is a major air pollutant gas produced by the combustion of fuels like coal. The main source of electricity production is by burning of fossil fuels in India and the whole world. The sulphur content of the coal varies from 1 to 4% and fortunately the Indian coal is low in sulphur content. SO₂ is also produced in the metallurgical operations.

(ii) **Oxides of nitrogen:** Oxides of nitrogen are produced either in the production of nitric acid or in the automobile exhausts and as the effluent of power plants. Out of the seven oxides of Nitrogen (N₂O, NO, NO₂, NO₃, N₂O₃, N₂O₄, N₂O₅) only nitric oxide and nitrogen dioxide are classified as the main pollutants. All the oxides of nitrogen are collectively known as NO_x.

(iii) **Carbon monoxide:** It is produced because of the incomplete combustion of coal and other petroleum products. It is produced in the exhaust of automobiles. In the pollution check of vehicles mainly CO and unburnt hydrocarbons are measured.

(iv) **Hydrogen sulphide:** Hydrogen Sulphide is an obnoxious (bad smelling) gas. It is produced mainly by the anaerobic (in absence of air) decomposition of organic matter. Other air polluting sulfur compounds are methyl mercaptan (CH₃SH) and dimethyl sulphide (CH₃-S-CH₃) etc.

(v) **Hydrogen fluoride:** It is an important pollutant even in very low concentrations. It is produced in the manufacturing of phosphate fertilizers.

(vi) **Chlorine and hydrogen chloride:** It is mixed in the air either from the leakages from water treatment plants or other industries where it is produced or used. Hydrogen chloride is also evolved in various industrial chemical processes. The main effect of chlorine is respiratory irritation which may be fatal.

(vii) **Ozone:** It is a desirable gas in the upper layers of the atmosphere as it absorbs the UV radiation of sunlight. But near the earth surface it is a poisonous gas. It makes poisonous chemicals by photochemical reactions.

(viii) **Aldehydes:** They are produced by the incomplete oxidation of motor fuels and lubricating oil. They may also be formed because of photochemical reactions. Formaldehyde is irritating to the eyes.

Classification according to chemical composition: (Organic – inorganic)

1. Sulfur-containing compounds.
2. Nitrogen-containing compounds.
3. Carbon-containing compounds.
4. Halogen-containing compounds.
5. Toxic substances (any of about).
6. Radiative compounds.

Classification according to physical state:

1. Gaseous.
2. Liquid (aqueous).
3. Solid.

Criteria air pollutants are six major pollutants defined by EPA (Environmental Protection Agency) for which ambient air standards have been set to protect human health and welfare. These include:

1. Ozone, O₃.
2. Carbon monoxide, CO.
3. Sulfur dioxide, SO₂.
4. Nitrogen oxides, NO_x.
5. Lead, Pb.
6. Particulates, PM₁₀.

Pollutant	Description	Sources	Health Effects	Welfare Effects
Carbon Monoxide (CO)	Colorless, odorless gas	Motor vehicle exhaust, indoor sources include kerosene or wood burning stoves.	Headaches, reduced mental alertness, heart attack, cardiovascular diseases, impaired fetal development, and death.	Contribute to the formation of smog.
Sulfur Dioxide (SO ₂)	Colorless gas that dissolves in water vapor to form acid, and interact with other gases and particles in the air.	Coal-fired power plants, petroleum refineries, manufacture of sulfuric acid and smelting of ores containing sulfur.	Eye irritation, wheezing, chest tightness, shortness of breath, lung damage.	Contribute to the formation of acid rain, visibility impairment, plant and water damage, aesthetic damage.
Nitrogen Dioxide (NO ₂)	Reddish brown, highly reactive gas.	Motor vehicles, electric utilities, and other industrial, commercial, and residential sources that burn fuels.	Susceptibility to respiratory infections, irritation of the lung and respiratory symptoms (e.g., cough, chest pain, difficulty breathing).	Contribute to the formation of smog, acid rain, water quality deterioration, global warming, and visibility impairment.
Ozone (O ₃)	Gaseous pollutant when it is formed in the troposphere.	Vehicle exhaust and certain other fumes. Formed from other air pollutants in the presence of sunlight.	Eye and throat irritation, coughing, respiratory tract problems, asthma, lung damage.	Plant and ecosystem damage.

Lead (Pb)	Metallic element	Metal refineries, lead smelters, battery manufacturers, iron and steel producers.	Anemia, high blood pressure, brain and kidney damage, neurological disorders, cancer, lowered IQ.	Affects animals and plants, affects aquatic ecosystems.
Particulate Matter (PM)	Very small particles of soot, dust, or other matter, including tiny droplets of liquids.	Diesel engines, power plants, industries, windblown dust, wood stoves.	Eye irritation, asthma, bronchitis, lung damage, cancer, heavy metal poisoning, cardiovascular effects.	Visibility impairment, atmospheric deposition, aesthetic damage.

Emission Sources

Major Classification of Air Pollution Sources:

1] Based on Origin: Natural and Manmade

While man-made air pollution does present health hazards, natural sources of air pollution can be equally dangerous at times. These sources include dust picked up by wind erosion, the emission of methane by livestock, and smoke from wildfires. Volcanic eruptions are perhaps the largest single source of air pollution, natural or man-made, that humans have ever dealt with. These can produce clouds of abrasive volcanic ash and other harmful substances such as chlorine and sulfur.

2] Based on Position: Stationary and Mobile

The sources of air pollution may be classified as stationary point sources (generally industrial in origin), diffuse or area sources and mobile sources (mainly cars and trucks).

3] Based on Axis of Release: Horizontal axis (Roadways traffic)

Vertical Axis release (Industrial Stacks)

4] Based on Intensity/frequency of release: Continuous release (Industrial Stacks)

Instantaneous release (Roadways traffic)

PHOTO CHEMICAL SMOG

Photochemical smog was first described in the 1950s. It is the chemical reaction of sunlight, nitrogen oxides and volatile organic compounds in the atmosphere, which leaves airborne particles and ground-level ozone. This noxious mixture of air pollutants can include Aldehydes, Nitrogen oxides, such as nitrogen dioxide, Peroxy acyl nitrates

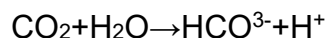
Tropospheric ozone, Volatile organic compounds etc. All of these chemicals are usually highly reactive and oxidizing. Photochemical smog is considered to be a problem of modern industrialization. It is present in all modern cities, but it is more common in cities with sunny, warm, dry climates and a large number of motor vehicles. Because it travels with the wind, it can affect sparsely populated areas as well.

VOCs + NO_x + sunlight → photochemical smog

ACID MIST/ RAIN

Definition

Normal Rain water pH is slightly acidic due to certain concentration of CO₂ dissolved as rainwater trickles down atmosphere.



Acid rain is defined as any type of precipitation with a pH that is unusually low or lower than 5.7.

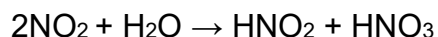
In the atmosphere, sulfur dioxide combines with water vapor to form hydrogen sulfite gas:
 $\text{SO}_2 + \text{H}_2\text{O} + \frac{1}{2} \text{O}_2 \rightarrow \text{H}_2\text{SO}_4$

Next, hydrogen sulfite reacts with oxygen to form sulfuric acid, a major component of acid rain:
 $\text{H}_2\text{SO}_3 + \frac{1}{2} \text{O}_2 \rightarrow \text{H}_2\text{SO}_4$

The sources of nitrogen oxides include the combustion of oil, coal and natural gas, forest fires, bacterial action in soil, volcanic gases, and lightning-induced atmospheric reactions.

In the atmosphere, nitrogen monoxide reacts with oxygen gas to form nitrogen dioxide gas:
 $\text{NO} + \frac{1}{2} \text{O}_2 \rightarrow \text{NO}_2$

Then, nitrogen dioxide reacts with water vapor in the atmosphere to form hydrogen nitrite and hydrogen nitrate:



Henceforth, acid rain is a mixture of HNO₃, H₂SO₄ + HCl. However, conditions needed to favor formation of these are sunlight, temperature, humidity, hydrocarbons, NO_x, SO₂.

OZONE DEPLETION

Ozone layer is an umbrella 24 km [15 miles] from earth surface, an essential component of the stratosphere that absorbs short wavelength ultraviolet radiation from the sun, heating the gases of the stratosphere in the process. World ozone day is celebrated on Sept, 16 of every year.

Stratospheric ozone is measured in Dobson units [DU] named after G.M.B Dobson who pioneered the study; [1 Dobson unit = 0.01 mm thickness of stratospheric ozone], Average ozone thickness in stratosphere is 300 DU, & when it falls below 200 DU, it's considered as Ozone hole. It is thinnest around equator and thickest near the poles.

Stratospheric ozone depletion is the term applied to the loss of stratospheric ozone molecules (O_3) and the disruption of Oxygen-Ozone concentration equilibrium in stratosphere [i.e., when chlorine atoms upset the natural O_2/O_3 equilibrium in the stratosphere]. Oxygen molecules interact with the intense solar radiation present at this elevation to form oxygen atoms. The oxygen atoms thus generated react with other oxygen molecules to form ozone(O_3).

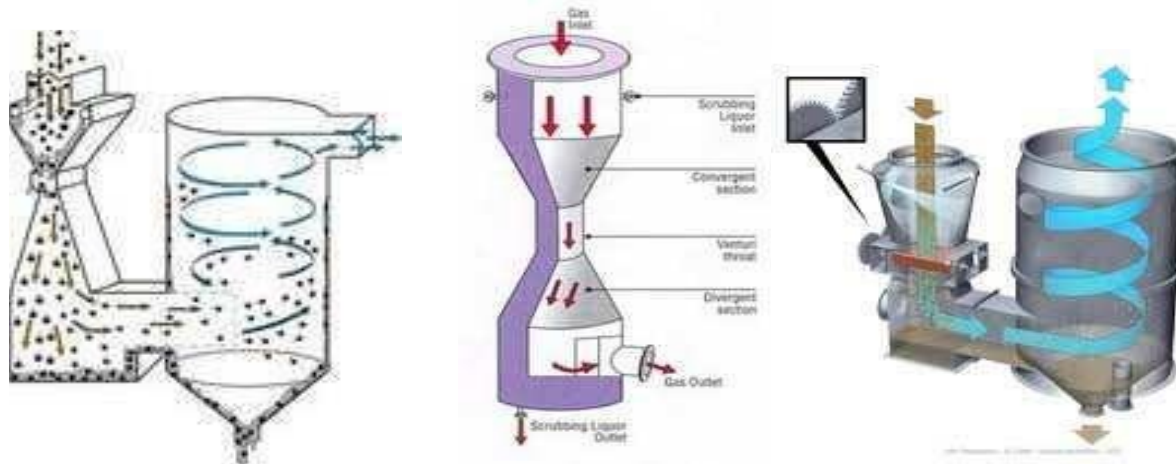
AIR POLLUTION CONTROL

CONTROL DEVICES FOR PARTICULATE EMISSIONS

Particulate matter is any finely divided liquid or solid substance. Examples of particulate matter include smoke, dust, or some forms of fine mist and is entrained in effluent gas streams or suspended in ambient air. Any particulate less than 10 micrometers (μm) in diameter is defined as PM10 and is regulated as a criteria pollutant. The purpose of such regulation is to control smaller, respirable particles that can bypass the body's respiratory filters and penetrate deeply into the lungs, which could cause harm to human health. Toxic substances, such as sulfates, sulfites, nitrates, heavy metals, and polycyclic organic matter are predominantly carried by particles in this size range. Therefore, control devices used today, to prevent particles from reaching the ambient air, focus on capturing particulate matter $\leq 10\mu m$ in diameter. Several factors must be considered when selecting an appropriate particulate control device. Typically, particles must be captured from an effluent gas stream; therefore, characteristics of the particles and the gas stream will determine the appropriate control device. Characteristics that must be considered include the particle size and resistivity, exhaust flow rate, temperature, moisture content, and various chemical properties of the exhaust stream such as explosiveness, acidity, alkalinity, and flammability.

VENTURI SCRUBBERS

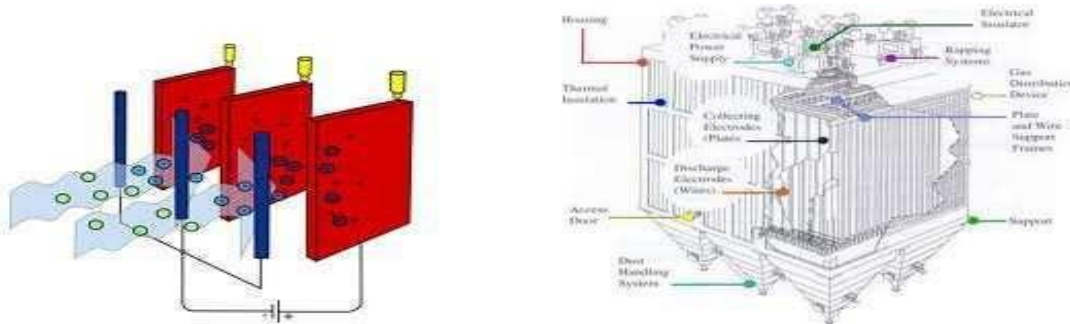
Venturi scrubbers use a liquid stream to remove solid particles. A venturi scrubber accelerates the waste gas stream to atomize the scrubbing liquid and to improve gas-liquid contact. In a venturi scrubber, a “throat” section is built into the duct that forces the gas stream to accelerate as the duct narrows and then expands. As the gas enters the venturi throat, both gas velocity and turbulence increase. Depending on the scrubber design, the scrubbing liquid is sprayed into the gas stream before the gas encounters the venturi throat, or in the throat, or upwards against the gas flow in the throat. The scrubbing liquid is then atomized into small droplets by the turbulence in the throat and droplet-particle interaction is increased. Some designs use supplemental hydraulically or pneumatically atomized sprays to augment droplet creation. However, the disadvantage of these designs is that clean liquid feed is required to avoid clogging. After the throat section, the mixture decelerates, and further impacts occur causing the droplets to agglomerate. Once the particles are captured by the liquid, the wetted PM and excess liquid droplets are then separated from the gas stream by an entrainment section, which usually consists of a cyclonic separator and/or a mist eliminator. Particle collection efficiencies of venturi scrubbers range from 70 to greater than 99 percent, depending on the application. Typically, venturi scrubbers are applied where it is necessary to obtain high collection efficiencies for fine PM. Thus, they are applicable to controlling emission sources with high concentrations of submicron PM.



ELECTROSTATIC PRECIPITATORS.

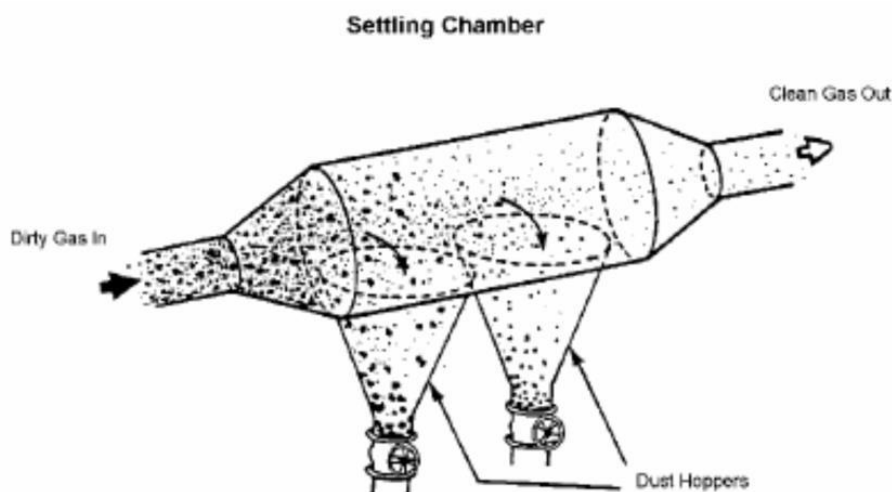
An ESP is a PM control device that uses electrical forces to move particles entrained within an exhaust stream onto collection surfaces. The entrained particles are given an electrical charge when they pass through a corona, a region where gaseous ions flow.

Electrodes in the center of the flow lane are maintained at high voltage and generate the electrical field that forces the particles to the collector plates. The high voltage electrodes are long wires or rigid “masts” suspended from a frame in the upper part of the ESP that run through the axis of each tube. Rigid electrodes are generally supported by both an upper and lower frame. The power supplies for the ESP convert the industrial AC voltage to pulsating DC voltage in the range of 20,000 to 100,000 volts as needed. The voltage applied to the electrodes causes the gas between the electrodes to break down electrically, an action known as a “corona.” The electrodes are usually given a negative polarity because a negative corona supports a higher voltage than does a positive corona



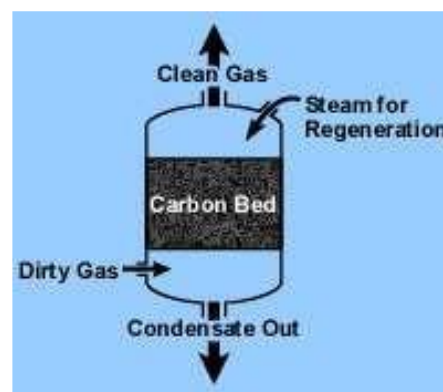
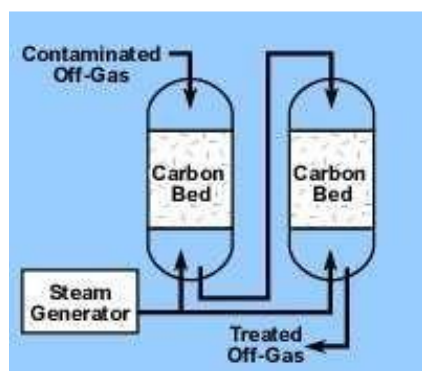
SETTLING CHAMBERS

This type of technology is a part of the group of air pollution controls collectively referred to as “pre-cleaners.” They are referred to as pre-cleaners because they are often used to reduce the inlet loading of particulate matter (PM) to downstream collection devices by removing larger, abrasive particles. Settling chambers are also referred to as gravity settling chambers, gravity collectors, expansion chambers, and outfall chambers. This is because settling chambers are quite effective in removing only large particles; therefore, they can be frequently used in combination with other control devices. Settling Chambers rely on simple gravitation to remove particles from a gas stream.



Control of gaseous pollutants from stationary sources - Adsorption

When a gas or vapor is brought into contact with a solid, part of it is taken up by the solid. The molecules that disappear from the gas either enter the inside of the solid, or remain on the outside attached to the surface. The former phenomenon is termed absorption (or dissolution) and the latter adsorption. Adsorption is the binding of molecules or particles to a surface. In this phenomenon molecules from a gas or liquid will be attached in a physical way to a surface. The binding to the surface is usually weak and reversible. The most common industrial adsorbents are activated carbon, silica gel, and alumina, because they have enormous surface areas per unit weight.



Air Pollutants

a) Sulphur dioxide (SO₂)

SO₂ is formed when fuel containing sulfur is burned. Sulfur is prevalent in raw materials such as crude oil, coal, and ore that contain common metals like aluminum, copper, zinc, lead etc. SO₂ reacts with other gases in the atmosphere to form sulphates that can cause harm to human health. Effects of SO₂ include respiratory illness, visibility impairment, acid rain and aesthetic damage. Sulfur oxides are emitted in significant quantities from thermal power plants, smelting process of sulfide ores to produce copper, lead and zinc and also from petroleum refining processes. The diesel driven vehicles are specific source of sulfur dioxide generated during combustion process. Sulfate particles, can be transported over long distances and deposited far from the sources. SO₂ can result in respiratory illness, particularly in children and the elderly, and it can also aggravate existing heart and lung diseases.

b) Oxides of Nitrogen (NO_x)

Oxides of nitrogen are a generic term for a group of highly reactive gases that contain nitrogen and oxygen in varying amounts. Nitrogen dioxide (NO₂) along with particulates is seen as a reddish brown layer over urban areas. Nitrogen oxides are formed when fuel

is burned at high temperature. Sources of nitrogen oxides includes vehicles, industrial processes that burn fuel. Oxides of nitrogen react with Volatile Organic Compounds (VOCs) to form ground level ozone. They also react to form nitrates, acid aerosols. They also contribute to nutrient overload that deteriorates water quality. Nitrogen dioxide irritates the nose and throat, and it appears to increase susceptibility to respiratory infections.

c) Particulate Matter (RSPM10 &PM2.5)

Particulate matter is a mixture of many subclasses of pollutants that contain many different chemical species. The particle size is often described by aerodynamic diameter. Aerodynamic diameter depends on particle density and is defined as the diameter of a particle with the same settling velocity as spherical particle with unit density i.e. 1 g/cm³ (USEPA, 1996). PM₁₀ are the particles with upper size limited by a 50% cut at 10 µm aerodynamic diameter (USEPA, 1996). PM₁₀ can be formed by physical processes of crushing, grinding and abrasion of surfaces. Mining and agricultural activities are some of the sources of large size particles. PM_{2.5} are the particles with upper size limited by a 50% cut at 2.5 µm aerodynamic diameter (USEPA, 1996). Particulate matter is called primary if it is in the same form chemical form in which it is emitted into the atmosphere. The primary particulate matter includes wind-blown dust such as road dust, fly ash, soot etc. Particulate matter is called secondary it is formed by chemical reactions in the atmosphere. Secondary particulate matter includes sulphates, nitrates, etc.

The size of particles is directly linked to their potential for causing health problems. Small particles less than 2.5 micrometers in diameter pose the greatest problems, because they can get deep into your lungs, and some may even get into your blood stream. Numerous scientific

studies have linked particle pollution exposure to a variety of problems, including increased respiratory symptoms, such as irritation of the airways, coughing, or difficulty breathing, decreased lung function, aggravated asthma, development of chronic bronchitis, irregular heartbeat, nonfatal heart attacks, and premature death in people with heart or lung disease (USEPA, 2008). People

with heart or lung diseases, children and older adults are the most likely to be affected by particle pollution exposure (USEPA, 2008). Environmental effects of particulate matter include visibility reduction, aesthetic damage etc.

Composition of Particulate Matter

Atmospheric particles include combustion-generated particles, such as diesel soot or fly ash; photochemically produced particles, such as those found in urban haze; and soil-like particles from resuspended dust. The major constituents of RSPM are organic and elemental carbon, metals/elements like silicon, magnesium, iron, ions like sulphates,

nitrates, ammonium etc. Understanding composition of particulate matter is most important to gain insight into the health effects caused and sources to be controlled. Composition of particulate matter varies from place to place and season depending upon sources present.

(i) Elemental Carbon

Elemental carbon (EC), also called black carbon“ or graphitic carbon“, has a chemical structure similar to impure graphite. Atmospheric elemental carbon is from primary anthropogenic sources and is not formed by reactions involving gaseous hydrocarbon precursors in the atmosphere. EC plays an important role in atmospheric chemistry because of its adsorptive and catalytic properties, which can capture other pollutants to react on its surface.

(ii) Organic Carbon

Organic carbon (OC), a mixture of hydrocarbons and oxygenates, is formed by a variety of processes, including combustion and secondary organic aerosol (SOA) formation. Organic carbon may be emitted as primary particles directly from sources, but secondary organics can also be formed in the atmosphere from the low-vapor-pressure products of atmospheric chemical reactions. OC is a complex mixture of different organic compounds, containing polycyclic aromatic hydrocarbons and other components.

(iii) Elements/Metals

Calcium, aluminum, silicon, magnesium, and iron are some of the crustal materials found predominantly in the coarse particles. Most of the elements are emitted from coal, oil combustion, vehicles, and industrial processes. Other sources include crustal material from road dust, tyre wear, construction activities, etc.

(iv) Ions

The common ions found in particulate matter are sodium, sulphates, nitrates, calcium, chloride, potassium. Potassium and nitrate may be found in both small size and coarse particles. Potassium comes from soil in coarse particles and in small size particles it comes from wood burning. Nitrate is formed by reaction of gas-phase nitric acid with gas-phase ammonia forming particulate ammonium nitrate.

d) Carbon Monoxide(CO)

Carbon monoxide is a colorless, odorless and poisonous gas. It is formed by incomplete combustion of carbon-containing fuels. Major source of CO are vehicles. Incomplete combustion is most likely to occur at low air-to-fuel ratios in the engine. These conditions are common during vehicle starting when air supply is restricted and are not tuned properly, and at altitude, where thin air effectively reduces the amount of oxygen available

for combustion. CO enters the bloodstream through lungs and forms carboxyhemoglobin which inhibits the blood's oxygen-carrying capacity to organs and tissues. Persons with heart disease are especially sensitive to carbon monoxide poisoning and may experience chest pain if they breathe the gas while exercising. Infants, elderly persons, and individuals with respiratory diseases are also particularly sensitive.

e) Ozone

Ozone is a secondary pollutant formed in the atmosphere by reaction between oxides of nitrogen and volatile organic compounds (VOCs) in the presence of sunlight. Vehicles, industrial emissions, gasoline vapours, chemical solvents, nitrogen oxides, and VOCs that form ozone. Peak O₃ levels typically occur during the warmer times of the year.

f) Ammonia

Ammonia is found in small quantities in the atmosphere, and is produced from the putrefaction of nitrogenous animal and vegetable matter. Ammonia occurs naturally and is produced by human activity. Ammonia and ammonium salts are also found in small quantities in rainwater. It is an important source of nitrogen which is needed by plants and animals. Ammonia gas can be dissolved in water and is called liquid ammonia or aqueous ammonia. Once exposed to open air, liquid ammonia quickly turns into a gas. Exposure to ammonia may occur by breathing or consuming food or water containing ammonia. No health effects have been found in humans exposed to typical environmental concentrations of ammonia. Exposure to high levels of ammonia in air may be irritating to skin, eyes, throat, and lungs and cause coughing and burns. Lung damage and death may occur after exposure to very high concentrations of ammonia.

g) Hazardous Air Pollutants

Hazardous air pollutants are also known as toxic air pollutants which may cause health effects such as reproductive effects, cancer etc. Toxic air pollutants include benzene, perchloroethylene, methylene chloride, dioxin, asbestos, toluene, and metals such as cadmium, mercury, chromium, and lead compounds. Sources of benzene are gasoline and perchlorethylene, is emitted from some dry cleaning facilities. Methylene chloride is used as a solvent and paint stripper by a number of industries. As per USEPA, 2007

people exposed to toxic air pollutants at sufficient concentrations and durations may have an increased chance of experiencing serious health effects and these health effects can include damage to the immune system, as well as neurological, reproductive (e.g., reduced fertility), developmental, respiratory, cancer and other health problems. Also as per USEPA, 2007, in addition to exposure from breathing air toxics, some toxic air pollutants such as mercury can deposit onto soils or surface waters, where they are taken up by plants and ingested by animals and are eventually magnified up through the food chain.

Air (Prevention and Control of Pollution) Act 1981

Government of India enacted the Air (Prevention and Control of Pollution) Act 1981 to arrest the deterioration in the air quality. The act prescribes various functions for the Central Pollution Control Board (CPCB) at the control level and State Pollution Control Boards at the state level. The main functions of the Central Pollution Control Board are as follows:

- To advise the Central Government on any matter concerning the improvement of the quality of the air and the prevention, control and abatement of air pollution.
- To plan and cause to be executed a nation-wide programme for the prevention, control and abatement of air pollution.
- To provide technical assistance and guidance to the State Pollution Control Board.
- To carry out and sponsor investigations and research related to prevention, control and abatement of air pollution.
- To collect, compile and publish technical and statistical data related to air pollution;and
- To lay down standards for the quality of air. The main functions of the State Pollution Control Boards are as follows:
- To plan a comprehensive programme for prevention, control and abatement of air pollution and to secure the execution thereof.
- To advise the State Government on any matter concerning prevention, control and abatement of air pollution.
- To collect and disseminate information related to air pollution.
- To collaborate with Central Pollution Control Board in programme related to prevention, control and abatement of air pollution;
- To inspect air pollution control areas, assess quality of air and to take steps for prevention, control and abatement of air pollution in such areas.

National Ambient Air Quality Standards(NAAQS)

The ambient air quality objectives/standards are a prerequisite for developing a programme for effective management of ambient air quality and to reduce the damaging effects of air pollution. The objectives of air quality standards are:

- To indicate the levels of air quality necessary with an adequate margin of safety to 7

Protect the public health, vegetation and property;

- To assist in establishing priorities for abatement and control of pollutant level;
- To provide uniform yardstick for assessing air quality at national level;and
- To indicate the need and extent of monitoring programme.

Air Quality Monitoring and AQI Considerations

The air quality monitoring network in India can be classified as (i) online and (ii) manual. The pollutant parameters, frequency of measurement and monitoring methodologies for two networks are very different. The AQI system for these networks could be at variance, especially for reporting and completeness in terms of parameters. (i) Online Monitoring network: These are automated air quality monitoring stations which record continuous hourly, monthly or annually averaged data. In India, ~ 40 automatic monitoring stations are operated (e.g. Figure 3.2: continuous stations in Delhi), where parameters like PM₁₀, PM_{2.5}, NO₂, SO₂, CO, O₃, etc. are monitored continuously. Data from these stations are available almost in real-time. Thus such networks are most suitable for computation of AQI sub-indices, as information on AQI can be generated in real time. For AQI to be more useful and effective, there is a need to set up more online monitoring stations for continuous and easy availability of air quality data for computation of AQI for more Indian cities.

Control Measures; Cost of Control Measures

Control measures for industrial facilities include adequate, well-designed, well-installed, efficiently operated and maintained air cleaning devices, also called separators or collectors. A separator or collector can be defined as an —apparatus for separating any one or more of the following from a gaseous medium in which they are suspended or mixed: solid particles (filter and dust separators), liquid particles (filter and droplet separator) and gases (gas purifier)II. The basic types of air pollution control equipment(discussed further in—Air pollution controlIII) are the following:

- For particulate matter: inertial separators (e.g., cyclones); fabric filters (baghouses); electrostatic precipitators; wet collectors(scrubbers)
- For gaseous pollutants: wet collectors (scrubbers); adsorption units (e.g., adsorption beds); afterburners, which can be direct-fired (thermal incineration) or catalytic (catalytic combustion).

Wet collectors (scrubbers) can be used to collect, at the same time, gaseous pollutants and particulate matter. Also, certain types of combustion devices can burn combustible gases and vapours as well as certain combustible aerosols. Depending on the type of effluent, one or a combination of more than one collector can be used.

The control of odours that are chemically identifiable relies on the control of the chemical agent(s) from which they emanate (e.g., by absorption, by incineration). However, when an odour is not defined chemically or the producing agent is found at extremely low levels, other techniques may be used, such as masking (by a stronger, more agreeable and harmless agent) or counteraction (by an additive which counteracts or partially neutralizes the offensive odour).

It should be kept in mind that adequate operation and maintenance are indispensable to ensure the expected efficiency from a collector. This should be ensured at the planning stage, both from the know-how and financial points of view. Energy requirements must not be overlooked. Whenever selecting an air cleaning device, not only the initial cost, but also operational and maintenance costs should be considered. Whenever dealing with high-toxicity pollutants, high efficiency should be ensured, as well as special procedures for maintenance and disposal of waste materials.

Control measures in vehicle emissions are adequate and efficient mandatory inspection and maintenance programmes which are enforced for the existing car fleet, programmes of enforcement of the use of catalytic converters in new cars, aggressive substitution of solar/battery-powered cars for fuel-powered ones, regulation of road traffic, and transportation and land use planning concepts.

Motor vehicle emissions are controlled by controlling emissions per vehicle mile travelled (VMT) and by controlling VMT itself (Walsh 1992). Emissions per VMT can be reduced by controlling vehicle performance - hardware, maintenance - for both new and in-use cars. Fuel composition of leaded gasoline may be controlled by reducing lead or sulphur content, which also has a beneficial effect on decreasing HC emissions from vehicles. Lowering the levels of sulphur in diesel fuel as a means to lower diesel particulate emission has the additional beneficial effect of increasing the potential for catalytic control of diesel particulate and organic HC emissions.

Reduction of VMT is an additional means of controlling vehicle emissions by control strategies such as

- use of more efficient transportation modes
- increasing the average number of passengers per car
- spreading congested peak traffic loads
- reducing travel demand.

Role of Environmental Impact Assessment

Environmental impact assessment (EIA) is the process of providing a detailed statement by the responsible agency on the environmental impact of a proposed action significantly affecting the quality of the human environment (Lee 1993).

EIA is an instrument of prevention aiming at consideration of the human environment at an early stage of the development of a programme or project. EIA is particularly important for countries which develop projects in the framework of economic reorientation and restructuring. EIA has become legislation in many developed countries and is now increasingly applied in developing countries and economies in transition.

EIA is integrative in the sense of comprehensive environmental planning and management considering the interactions between different environmental media. On the other hand, EIA integrates the estimation of environmental consequences into the planning process and thereby becomes an instrument of sustainable development. EIA also combines technical and participative properties as it collects, analyses and applies scientific and technical data with consideration of quality control and quality assurance, and stresses the importance of consultations prior to licensing procedures between environmental agencies and the public which could be affected by particular projects. A clean air implementation plan can be considered as a part of the EIA procedure with reference to the air.

AIR QUALITY MONITORING

Air quality monitoring means the systematic measurement of ambient air pollutants in order to be able to assess the exposure of vulnerable receptors (e.g., people, animals, plants and art works) on the basis of standards and guidelines derived from observed effects, and/or to establish the source of the air pollution (causal analysis). 19 Ambient air pollutant concentrations are influenced by the spatial or time variance of emissions of hazardous substances and the dynamics of their dispersion in the air. As consequence, marked daily and annual variations of concentrations occur. It is practically impossible to determine in a unified way all these different variations of air quality (in statistical language, the population of air quality states). Thus, ambient air pollutant concentrations measurements always have the character of random spatial or time samples.

Measurement procedures for dust-forming air pollutants and its composition

Among particulate air pollutants, dustfall and suspended particulate matter (SPM) are differentiated. Dustfall consists of larger particles, which sink to the ground because of their size and thickness. SPM includes the particle fraction that is dispersed in the atmosphere in a quasi-stable and quasi-homogeneous manner and therefore remains suspended for a certain time. *Measurement of suspended particulate matter and metallic compounds in SPMA*s is the case with measurements of gaseous air pollutants, continuous and discontinuous measurement procedures for SPM can be differentiated. As a rule, SPM is first separated on glassfibre or membrane filters. It follows a gravimetric or radiometric determination. Depending on the sampling, a distinction can be made between a procedure to measure the total SPM without fractionation according to the size of the particles and a fractionation procedure to measure the fine dust. The advantages and disadvantages of fractionated suspended dust measurements are disputed internationally. In Germany, for example, all threshold limits and assessment standards are based on total suspended particulates. This means that, for the most part, only total SPM measurements are performed. In the United States, on the contrary, the so-called PM-10 procedure (particulate matter $\leq 10\mu\text{m}$) is very common. In this procedure, only particles with an aerodynamic diameter up to 10 μm are included (50 per cent inclusion portion), which are inhalable and can enter the lungs. The plan is to introduce the PM-10 procedure into the European Union as a reference procedure. The cost for fractionated SPM measurements is considerably higher than for measuring total suspended dust, because the measuring devices must be fitted with special, expensively constructed sampling heads that require costly maintenance. Table contains details on the most important SPM measurement procedures.